Unil.
HEC Lausanne

Applied Investments

Fall 2026 – Semester 5.1
Amit Goyal

Course Description

This course is designed to acquaint students with the state-of-the-art quantitative equity investing employed in the asset management industry. Students targeting asset management careers or planning to take the CFA tests will find this course particularly useful. The students will build quantitative portfolio strategies to identify alpha opportunities.

Course Grading

The final grade is composed of

1. Paper presentation (30%)
2. Class participation (10%)
3. Portfolio strategy project (60%)

More details on these components are provided later. The retake will take place according to HEC/UNIL guidelines.

Course Material

Class materials (copies of slides, excel spreadsheets, and other relevant readings) will be posted on the Moodle website https://moodle.unil.ch/course/view.php?id=37736. I encourage you to check this website regularly. The following textbooks are not required but helpful:

- Andrew Ang, 2014, *Asset Management: A Systematic Approach to Factor Investing*, Oxford University Press.
- Turan G. Bali, Robert Engle, and Scott Murray, 2016, *Empirical Asset Pricing: The Cross Section of Stock Returns*, Wiley-Blackwell.
- Ludwig B. Chincarini and Daehwan Kim, 2022, *Quantitative Equity Portfolio Management*, 2nd ed, McGraw Hill.
- Richard C. Grinold and Ronald N. Kahn, 1999, *Active Portfolio Management*, McGraw Hill.
- Lasse Heje Pedersen, 2015, *Efficiently Inefficient*, Princeton University Press.
- Edward E. Qian, Ronald H. Hua, and Eric H. Sorensen, 2007, *Quantitative Equity Portfolio Management*, Chapman & Hall.
- Thierry Roncalli, 2014, *Introduction to Risk Parity and Budgeting*, Chapman & Hall.
- Bernd Scherer, 2010, *Portfolio Construction and Risk Budgeting*, 4th ed, Risk Books.
- Christophe Scheuch, Stefan Voigt, Patrick Weiss, and Christoph Frey, 2024, *Tidy Finance with Python*, Chapman and Hall/CRC, https://www.tidy-finance.org/python/.

There is no teaching assistant for the class.

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Course Outline

Our class meets weekly from 14:15 to 18:00 on Mondays in Internef/122.

The following is the tentative schedule but remains subject to change (and will be dynamically adjusted).

No.	Date	Topic
01	14 Sep	Introduction/Portfolio strategies
02	21 Sep	— PUBLIC HOLIDAY—
03	28 Sep	Factor models
04	05 Oct	Trading costs
05	12 Oct	Aggregate predictability
06	19 Oct	Paper presentations (1)
07	26 Oct	Paper presentations (2)
08	02 Nov	Paper presentations (3)
09	09 Nov	Paper presentations (4)
10	16 Nov	Group projects (1)
11	23 Nov	Group projects (2)
12	30 Nov	— NO CLASS —
13	07 Dec	Group projects (3)
14	14 Dec	Group projects (4)

Paper presentations

You will work in groups of 1-3 students. You are free to choose your own group members.

Choose one paper from the following list and sign up on

https://docs.google.com/spreadsheets/d/19m684Zb0aSZ9LKOefvNcxKXHPe0G4ngBygoaT97ySko/edit?usp=sharing

You will prepare a 15-minute presentation to be made during class. In preparing your presentation:

- Present the main research question and results of the paper.
- Use resources other than the article itself. This can be other papers cited by the paper, other papers that cite the paper, newspaper articles, interviews, movies, your own related trading experience, your own empirical exercise using different data samples, etc.

In general, taking a critical view of the results will be more highly rewarded than merely regurgitating the results in the paper.

The list of papers and the date on which they will be presented is in Google Spreadsheet. All the papers can be found below:

https://www.dropbox.com/scl/fo/l2hq5xaucp8au6uc4kag6/AKEKD2pqwW9GXVPyAMun4Sw?rlkey=vhhbfeuexyuynfb8c7ktp77gc&st=m936jf0e&dl=0

Portfolio strategy project

You will work in groups of 3-5 students. You are free to choose your own group members.

Sign for groups at
https://docs.google.com/spreadsheets/d/1vwiJd5tsMNNAMqVnFNxP5tgx9e_43Ar18X0qpqBVTTs/edit?usp=sharing

From a fund manager's perspective, we would like to know if an investment strategy has been "worthwhile." Merely picking stocks ignores the benefits of diversification and carries a lot of idiosyncratic risk. Yet the theoretical result of Modern Portfolio Theory does not hold either – there are possibly exploitable anomalies in the risk-return relationship. The market may not be efficient.

You will construct your own innovative strategy. This project involves two major steps:

1. Data
Any empirical analysis of investment will require data. You will have to collect and describe return data on the assets you would want to include in your analysis of a portfolio. Since the whole class revolves around equity strategies, we will restrict ourselves to equity data. And since the data on the US market are plentiful and easy to retrieve, we will further restrict ourselves to the US market data. If you need to deviate from these restrictions, please seek pre-approval.

2. Constructing trading strategies
Use the data to construct your strategy. You will need to decide which metrics to use in evaluating your strategy. For example, you may want to compute alphas of your strategy to popular asset pricing models.

This assignment can be seen as a (small) scientific paper. The length of the paper (excluding tables, graphs and references) should be no more than 20 pages (1.5 line-spaced, 12pt, 1-inch margins).

Submission deadline: 23h59, Friday, 13 November 2026 (for all groups).

You will also present the results of your strategy in class. Prepare a 15-minute presentation assuming that you are presenting to the investment committee of your asset management firm. You are free to decide how best to present your results.

The exact date for your group's presentation will be announced later.